

# Devansh Parapalli

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## Professional Summary

AI/ML and backend systems engineer with end-to-end production ownership across LLM inference infrastructure, agentic pipelines, event-driven microservices, and container-orchestrated platforms. Founding engineer and technical lead on Cisco's Finance agent platform — from requirements through production operations. Achieved MRR 0.75 / nDCG@5 0.7 over 100K-chunk retrieval corpus. Cut video processing time 12x and case resolution time from 1+ day to under 30 seconds. Published AI/ML researcher (IJTE, 2025). GCP Professional Cloud Architect certified.

## Work Experience

### Systems Engineer

Tata Consultancy Services Limited

Jul 2025 — Present

Hyderabad, India

- **Client: Cisco Capital** — Founding engineer; scaled the team to 8 engineers under my technical lead within 6 months, owning requirements, architecture, and cross-functional alignment with support, FinOps, and audit. Outcomes reviewed directly by client stakeholders.
  - Reduced case resolution time from **1+ day to under 30 seconds** for **750+ monthly tickets** at **95% autonomous closure** (4% pending human verification, 1% escalated) by building **AIT CaseIQ**: designed multi-intent detection, agentic tool-calling LLM workflows, and structured output delivery over a **Kafka**-backed event-driven ingestion architecture with auto-scaling containerized runners on OpenShift/Kubernetes, decoupling load spikes from processing capacity.
  - Built the agent framework on **Google ADK** with backward-compatible **A2A** protocol extensions and a codegen layer that lets business users stand up agents themselves; runs **6 tracks** and **8 agents**, all shipped within the first month of codegen; a new track onboards in **4–5 hours**.
  - Automated SOX audit review eliminating **1,000+ person-hours/month** of payroll reconciliation by engineering a Python pipeline that diffs PAAT payroll records against Oracle General Ledger accruals, feeds outstanding variances and incoming email justifications into an LLM scoring layer, and tracks gaps — replacing the manual auditor review loop end-to-end.
  - Saved **1,300+ person-hours/month** across **4 FinOps workflows** by building async LangGraph-orchestrated pipelines with idempotent retry logic: an SOP-driven multi-source report generator (Snowflake, MongoDB, OracleDB), a RAG + tool-calling knowledge chatbot, a natural language to multi-target query translator (SQL + NoSQL), and an automated data aggregation and Excel delivery pipeline.
  - Built a data reconciliation engine that compiles plain-English steps into processing and matching operations, running **100+ workflows** at **1M+ rows per side** with type-aware normalization that eliminates false positives.
  - Built an independent audit layer for a production document-processing pipeline, using a deliberately different methodology so agreement between the two is real evidence; catches a real error in ~1 of 30 documents the primary pipeline passed at a **1-in-1,000** false-flag rate, sampling **30% of global invoices** daily. Calibrated with eval sets and human review; replaced an LLM-as-a-Judge prototype on speed and cost.
  - Led the AI reskilling of **~70 application-support engineers and managers** into AI-native workflows; **75%** now run AI agents in their day-to-day work.
- **Internal: GenAI Centre of Excellence** — Led technical architecture across **14 RFPs**, **7 won** totalling **\$450M+ TCV**, delivering technical architecture documents and working prototypes demoed directly to prospective clients, covering GenAI enablement, cloud migration, high-availability architecture, and legacy infrastructure modernization.
  - Eliminated external API dependency for enterprise model serving (**7B–120B parameters**) by architecting on-premises **vLLM + LiteLLM** inference platform with centralized API gateway routing, RBAC, PII sanitization, and audit logging.
  - Authored a 21-use-case GenAI adoption roadmap for a European semiconductor manufacturer on Vertex AI and Gemini Enterprise, including on-premises infrastructure that keeps confidential engineering data inside their data center.
  - Designed AI use-cases for a global telecom equipment maker (migration of SAP ME manufacturing software to GKE across all production sites, with an Assess-Migrate-Manage workflow), a barcode and mobile computing hardware maker (large-scale PO/invoice ingestion with multi-source merging for handwritten, scanned and digital documents), and a print and document technology company (proactive machine health monitoring, intelligent ticket assistant, automated patch testing).

### GenAI & Backend Development Intern

PolymathAI

Jun 2024 — Aug 2024

Nagpur, India

- Cut per-video AI processing time **12x (3 min → 15 sec)** by designing a parallelized inference pipeline across horizontally-scaled containers, processing **14,600+ videos** while optimizing per-instance request caps and cold-start overhead for cost efficiency. System extracted frames at 1fps, processed images through separate inference calls, and merged results for flashcard generation, Q&A, and natural language search.
- Scaled platform to support **100 concurrent users**; built complete pipeline for video bite creation and multi-modal content understanding enabling interactive educational content generation.
- Led full-stack development of web extensions using MERN stack and Next.js; developed robust content scraping framework supporting multiple data formats.

- Integrated Razorpay and Stripe payment gateways, reducing checkout friction by **80%** and increasing user conversion by **35%**.
- Implemented e-commerce features including discount code engine, shopping carts, and multi-processor payment integration using React, Express, MongoDB, and TailwindCSS; resolved critical inventory management and order processing bottlenecks. Integrated Firebase authentication; executed game request feature.

**Independent Technical Consultant**  
Freelance Engagements

2020 — 2024  
Remote

- Delivered 10+ complete software solutions across web applications, mobile development, automation systems, and data migration projects — from requirements gathering through deployment.
- Built diverse production systems: landing pages with Three.js animations, inventory management systems, e-commerce platforms, web scraping/automation workflows, and RAG-powered conversational interfaces. Stack: React, Svelte, Node.js, Express, Python, Selenium, LangChain, vLLM, Ollama, cloud LLM APIs.
- Identified security vulnerabilities through reverse engineering, earning bug bounties for critical findings in 3 Android applications and 2 web platforms.

## Projects

**AIKO: AI-powered Knowledge Organizer** ([aiko.parapalli.dev](https://aiko.parapalli.dev))

Jun 2024 — Present

- Achieved **MRR 0.75** and **nDCG@5 0.7** over **100K-chunk technical knowledge corpus** by implementing hybrid retrieval combining pgvector semantic search with keyword fallback, tuned chunk strategy, and embedding model selection across multi-modal content (PDFs, web sources, audio, images). Published in IJTE, March 2025.
- Sustained **p95 ingest latency of 3s** and **p99 of 10s** across multi-modal pipeline via custom serialization layer with cloud-native and offline-capable deployment targets.
- Engineered custom NLP pipeline reducing average input prompt size from 8,192 to 4,608 tokens (**44% reduction**) using embedding-based retrieval and contextual injection, validated via cosine + manhattan distance similarity scoring with LLM verification — directly reducing API costs and improving response latency.
- Reduced LLM response latency by **25%** by replacing SSE with custom **WebSocket** duplex streaming for real-time token delivery.
- Designed novel single-file plugin architecture eliminating traditional plugin boilerplate; **90% code reduction** vs LangChain plugin patterns. Implemented 90%+ test coverage and automated CI/CD via GitHub Actions.
- Expanded ingestion surface to all major browsers via cross-browser **WebExtension** with client-side content capture and real-time backend synchronization.

**ARA: AI-powered Research Assistant** ([ara.parapalli.dev](https://ara.parapalli.dev))

Jan 2024 — May 2024

- Built full-stack RAG-based research workspace with inline web retrieval and per-prompt source citation as sole engineer across SvelteKit, FastAPI, Supabase, and containerized LLM runners; deployed across Vercel, Railway, and Modal.
- Reduced literature review time by **60%**; achieved **80% faster** mean LLM response times vs industry benchmarks using custom WebSocket streaming over SSE.
- Re-implemented core dataflow using C-based Python libraries for tokenization and embedding; re-implemented LangChain in pure Python for **50% additional latency reduction**.
- Published: ResearchGate DOI: <https://doi.org/10.13140/RG.2.2.28162.90567/1>

**Adhyaaya'23 — College Tech-Fest Web App** ([adhyaaya23.parapalli.dev](https://adhyaaya23.parapalli.dev))

Dec 2022 — Mar 2023

- Spearheaded full-stack web app for college techfest, increasing event registration efficiency by **70%**.
- Integrated Razorpay gateway reducing checkout friction by **65%**; achieved **40% improvement in page load times** via Svelte, SvelteKit, Three.js, TailwindCSS frontend. Backend in Deno + Python; max TTFB 0.13s. Database schema in Supabase with **55% query performance improvement**.

**Automatic Academic Result Extraction & Analysis** ([github.com/DevParapalli/rtnnu-result-scraping](https://github.com/DevParapalli/rtnnu-result-scraping))

May 2022 — Present

- Engineered resilient web scraper handling site inconsistencies with **99.8% accuracy**, parsing PDF admit cards and extracting structured results for 1000+ students.
- Designed ranking algorithms providing previously-unavailable college and university-wide performance insights; now used by **5 departments** for academic planning.
- Implemented browser-application context synchronization for checkpoint-based recovery.

## Education

**Government College of Engineering, Nagpur**

Nagpur, Maharashtra, India

*Bachelor of Technology, Computer Science and Engineering*

Dec 2021 — Jun 2025

- CGPA: 8.28/10.0 | Notable Project: Development Containers and CI/CD Pipelines across languages and frameworks.

**G. H. Raisoni Vidyapeeth**

Nagpur, Maharashtra, India

*12th Standard | CBSE*

Apr 2019 — Mar 2021

- Percentage: 85/100

## Technical Skills

- **AI/ML Engineering:** LLM Inference (vLLM, LiteLLM, Ollama), Tool Calling, MCP, Agentic AI, LangGraph, Multi-Agent Workflows, Hybrid Retrieval, Chunk Strategy, Transformers, Hugging Face, Intent Detection, Semantic Ranking, Structured Output Parsing, RAG architectures, pgvector. High proficiency with LangChain; experience with Azure OpenAI, GCP Vertex AI, AWS Bedrock, Anthropic, Mistral, Cohere.
- **Languages & Frameworks:** Python (5+ yrs, expert), JavaScript/TypeScript (4+ yrs, expert), C/C++ (3+ yrs), SQL, Zig, Rust, Java. FastAPI, Django, Node.js/Bun, React.js, Next.js, Svelte/SvelteKit, Express, TailwindCSS, Three.js, PWA.
- **Distributed Systems:** Kafka, Event-Driven Architecture, Microservices, Horizontal Scaling, API Gateways, WebSockets.
- **Databases:** PostgreSQL (w/ pgvector), Redis, MongoDB, OracleDB, Snowflake, SQLite, Spanner, Supabase, Firebase.
- **Cloud & Infrastructure:** GCP (Certified Professional Cloud Architect — Document AI, Vertex AI, Cloud Run, Compute Engine), Docker, Kubernetes, OpenShift, Terraform.
- **DevOps & Tools:** GitHub Actions CI/CD, Jenkins (Jenkinsfile), SonarQube, Git, Linux system administration, web scraping (Selenium), process automation.
- **Embedded & Other:** IoT (Arduino, ESP32, Raspberry Pi), WebGL, Robotics programming, CUDA (NVIDIA certified).

## Leadership & Technical Service

**Vice President, Computer Science Students' Association (COMSA)** Sep 2023 — Aug 2024

- Managed team of 20 student leaders organizing departmental technical festivals, hackathons, and workshops. Launched skill development initiatives resulting in **500+ technical certifications** earned by students across the college.

**Head, E-Yantra Robotics Club** Jan 2023 — May 2025

- Revitalized dormant club from **2 members to 45+ active participants** through strategic recruitment, workshops, and competition-focused projects. Mentored students in embedded systems, sensor integration, and autonomous navigation; coordinated national robotics competitions. Established technical curriculum leveraging WRO international experience.

**Co-Head Student Coordinator, Institution's Innovation Council (IIC 5.0)** Sep 2022 — Aug 2023

- Organized innovation-focused events, mentored student technical projects, established internship pathways, and guided teams through project development lifecycles.

**Technical Lead, Google Developer Student Club** Aug 2022 — Jul 2023

- Managed technical infrastructure for student developer community; organized workshops on modern development practices; mentored students on cloud technologies and software engineering fundamentals.

**Technical Head, Student Welfare Committee (SWC) / Gymkhana** Sep 2022 — Aug 2023

## Competitive Achievements

**Smart India Hackathon 2023 — National Finalist** Dec 2023

- Led AI development for exam proctoring system detecting cheating via computer vision: test-taker positioning, gaze direction, unauthorized secondary display detection. Architected locally-running privacy-focused alternative to cloud proctoring.

**IDE Bootcamp (AICTE, for SIH Finalists) — Winner** 2024

**World Robotics Olympiad — UAE Runner-up (2nd Place, Open Category)** Nov 2017

- Designed and programmed autonomous farming robot using Lego Mindstorms + Raspberry Pi: field navigation, soil preparation, seed planting, watering, video monitoring. Responsible for RPi control system and sensor integration. 2nd place, Abu Dhabi international competition.

**TCS ITWiz — Regional Finalist** 2018, 2019

**GitHub Archive Program — Contributor** 2020

**Modi-Script Open Source Project — Contributor** 2020

## Publications

**Artificial Intelligence based Knowledge Organizer for diverse data formats** IJTE – ISTE India  
• Vol. 48, Special Issue No. 2, Page 261–265 March 2025

**ARA: AI-powered Research Assistant** ResearchGate  
• <https://doi.org/10.13140/RG.2.2.28162.90567/1> April 2024

## Certifications

All Google Cloud credentials are verifiable at [credly.com/users/devansh-parapalli](https://credly.com/users/devansh-parapalli).

**Professional Cloud Architect** Dec 2025 — Dec 2027  
Credential ID: e9709e75c5fb4491b383e0be91106499 Google Cloud Platform

**Cloud Digital Leader** Sep 2025 — Sep 2028  
Credential ID: 247dd2671d194d66b3466a1702709631 Google Cloud Platform

**Generative AI Leader**

Credential ID: c6d67c50865d41dab9a32452654340ee

Sep 2025 — Sep 2028

Google Cloud Platform

**Fundamentals of Accelerated Computing with CUDA C/C++**

Certificate ID: 4FSGnu7QTPqoyVYAZHLiUw

Jul 2024

NVIDIA